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STRUCTURAL ENGINEERING – OCEAN TRANSPORTATION ANALYSIS FOR MODULES

04 – Definition of Axis Orientations and Positive Directions

This attachment describes the coordinate systems that apply to module marine transportation. There are two systems that apply:

- Section 1: LTV system used for the marine vessel.
- Section 2: XYZ system used by RISA-3D.

Translation of coordinates, from LTV to XYZ are described in Section 3.

1.0 THE LTV COORDINATE SYSTEM USED FOR THE MARINE VESSEL

Used to input marine vessel properties as well as translation and rotation acceleration parameters.

A right-handed Cartesian coordinate system (L, V, T) is defined as follows when facing the direction of travel:

- The L axis (longitudinal) is horizontal and positive forward
- The T axis (transverse) is horizontal and positive to the left or port side
- The V axis (vertical) is vertical and positive upward

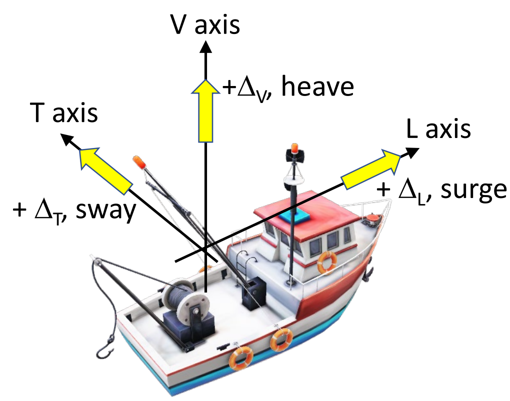

Figure 1: Positive LTV System Translations (Δ_L, Δ_V Δ_T)

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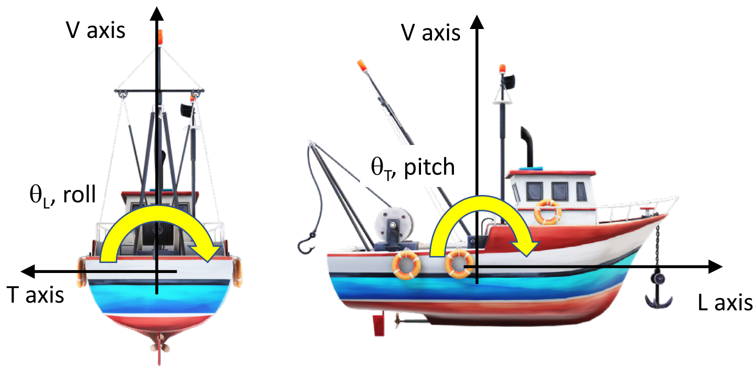

Figure 2: Positive LTV System Rotations (θ_L, θ_V, θ_T)

2.0 THE XYZ COORDINATE SYSTEM USED BY RISA-3D

A right-handed XYZ Cartesian coordinate system (X, Y, Z) is defined with an upward +Y axis.

With respect to this document, any orientation may be used provided that the Y axis is positive upward. However, to facilitate uniform communications and reporting, it is recommended to standardize the jobsite coordination system for a project.

Suggested Orientation:

On each module, the +X axis should be consistently positive to the east (shown in Figure 3) or consistently positive to the north. This setting should match the positive directions in the pipe stress reports.

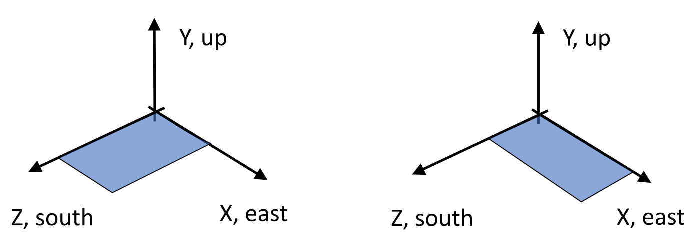

Figure 3: Positive XYZ System Orientation

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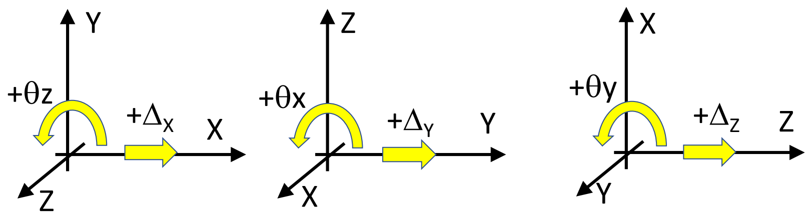
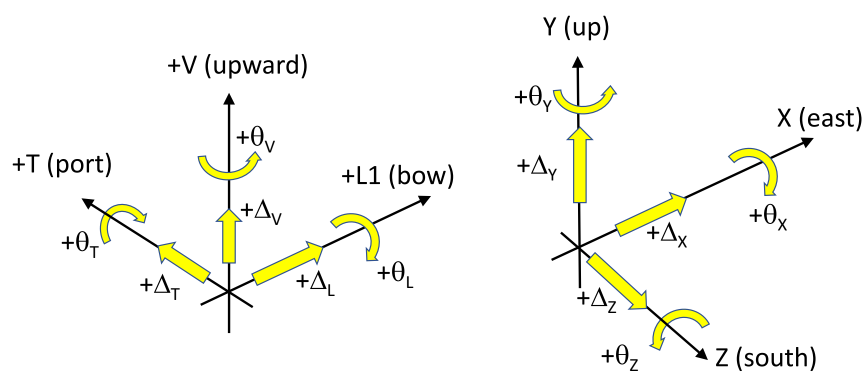


Figure 4: Positive XYZ System Positive Translations & Rotations

3.0 TRANSLATING FROM LTV SYSTEM TO XYZ SYSTEM

1) Direction of travel is in the +X axis:



$+\Delta_L$ translates to $+\Delta_X$
$+\Delta_T$ translates to $-\Delta_Z$
$+\Delta_V$ maps to $+\Delta_Y$

$+\theta_L$ translates to $+\theta_X$
$+\theta_T$ translates to $-\theta_Z$
$+\theta_V$ translates to $+\theta_Y$

Figure 5: Positive X Axis Aligned with Travel Direction

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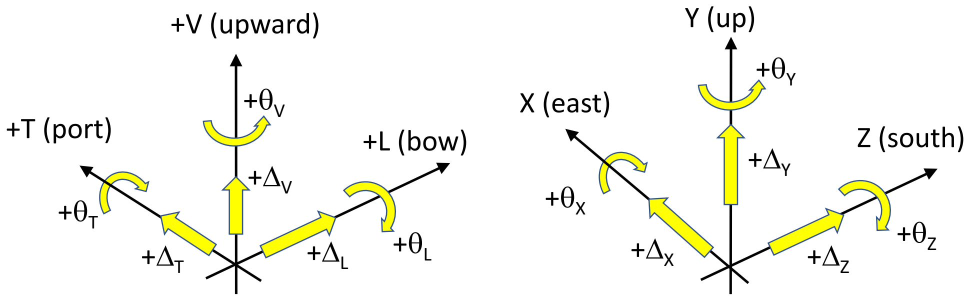
2) Direction of travel is in the +Z axis:



$+\Delta_L$ translates to $+\Delta_Z$	$+\theta_L$ translates to $+\theta_Z$
$+\Delta_T$ translates to $+\Delta_X$	$+\theta_T$ translates to $+\theta_X$
$+\Delta_V$ translates to $+\Delta_Y$	$+\theta_V$ translates to $+\theta_Y$

Figure 6: Positive Z Axis Aligned with Travel Direction